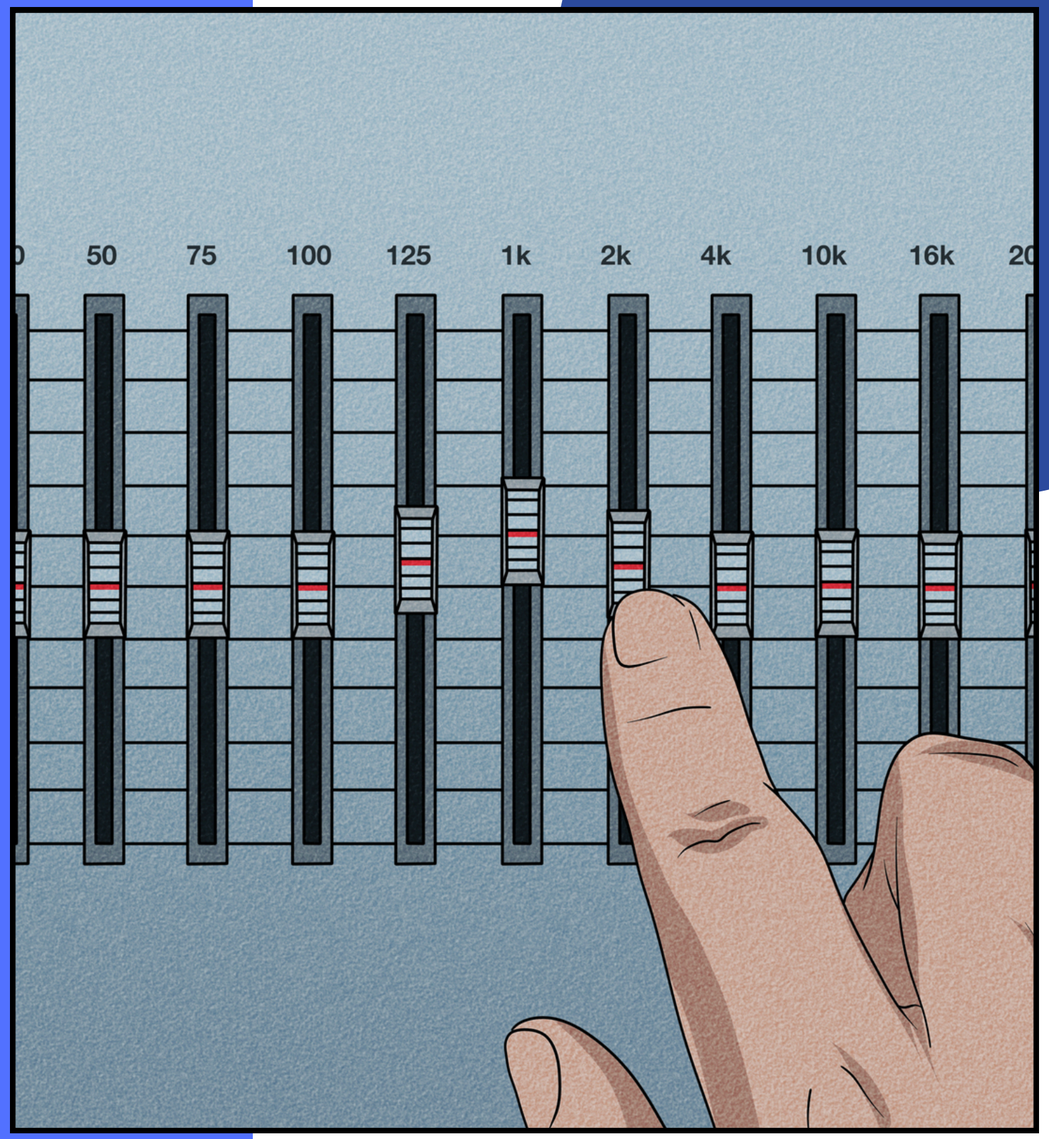
# ECUALIZADOR
# BASADO EN FILTROS DIGITALES

# INTRODUCCIÓN

Los ecualizadores digitales son herramientas esenciales en el procesamiento de señales de audio, permitiendo ajustar selectivamente las ganancias de diferentes bandas de frecuencia. Este proyecto combina filtros FIR e IIR para desarrollar un sistema versátil y práctico.



# ¿Qué son filtros digitales?

Son sistemas que procesan señales discretas para modificar su contenido frecuencial.

Tienen como objetivo:

- Ajustar ganancias en bandas específicas
- Eliminar ruido o interferencias
- Mejorar calidad de audio

**Ecuación**

$$y[n] = \sum_{k=0}^{N} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$

Donde:

- N = orden del filtro
- $a_k$ = coeficientes de realimentación
- $b_k$ = coeficientes de alimentación directa

# Tipos de Filtros

## FIR

Estructura no recursiva
Ecuación:

$$y[n] = \sum_{k=0}^{N} b_k x[n-k]$$

Ventajas:

- Estabilidad garantizada
- Fase lineal (ideal para audio)

Desventaja:

- Alto costo computacional (requiere órdenes altos)

## IIR

Estructura recursiva
Ecuación:

$$y[n] = \sum_{k=0}^{N} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$

Ventajas:

- Eficiencia (menor orden = menos recursos)
- Respuesta frecuencial más abrupta

Desventaja:

- Riesgo de inestabilidad (polos fuera del círculo unitario)

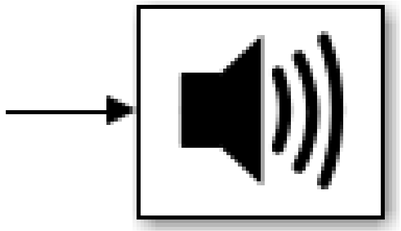
# Implematacion Simulink

MATLAB® & SIMULINK®

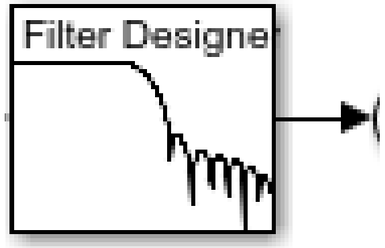
# BLOQUES

From Multimedia File

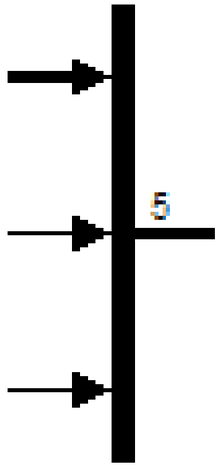
Gain3



Audio Decive Writer



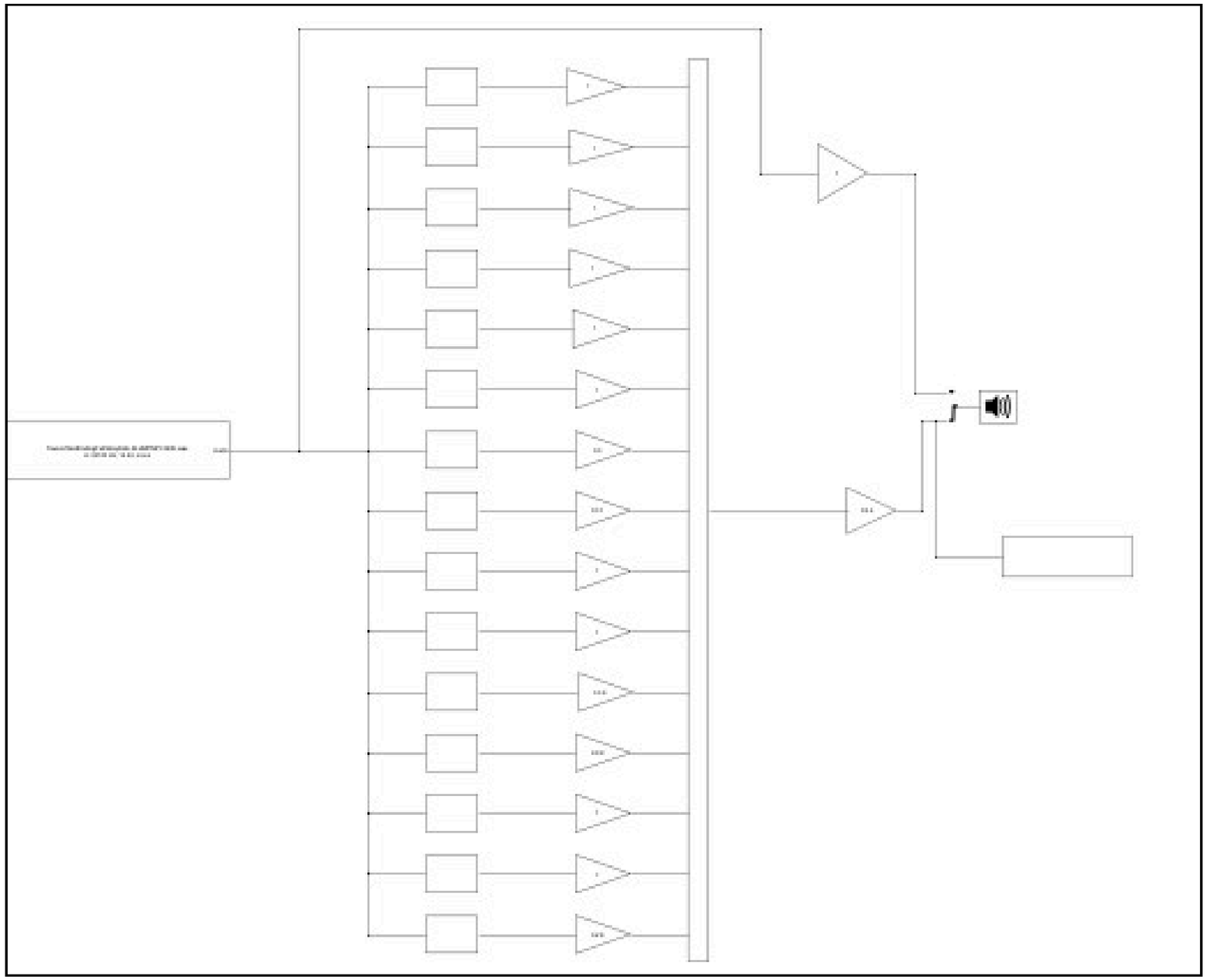

Digital
Filter Design



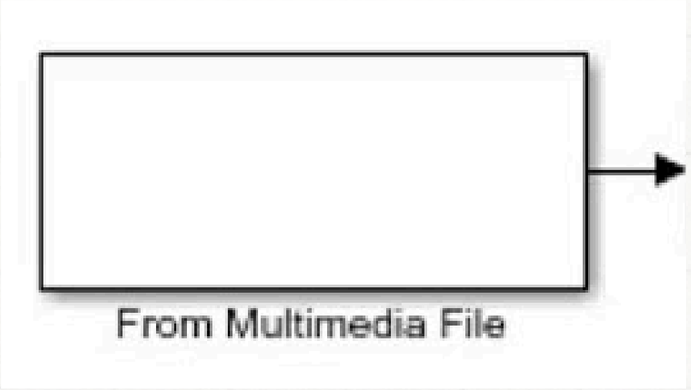

Mux

To Multimedia File

# ESQUEMA GENERAL DE SIMULINK



# CONFIGURACIÓN




Block Parameters: From Multimedia File

From Multimedia File

Reads multimedia files containing audio, video, or audio and video data.

For Windows platforms, this block reads compressed or uncompressed multimedia files.

For non- Windows platforms, this block reads uncompressed video and audio AVI files, and video only, compressed or uncompressed files.

Main | Data Types

Parameters

File name: ghtmare (Full Version)(MP3_128K).mp3 Browse...

☑ Inherit sample time from file

Number of times to play file: inf

Read range: [1 Inf] *[1,Inf]*

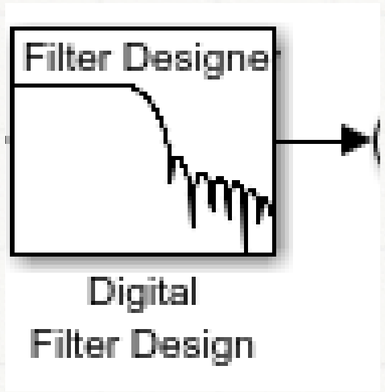
Outputs

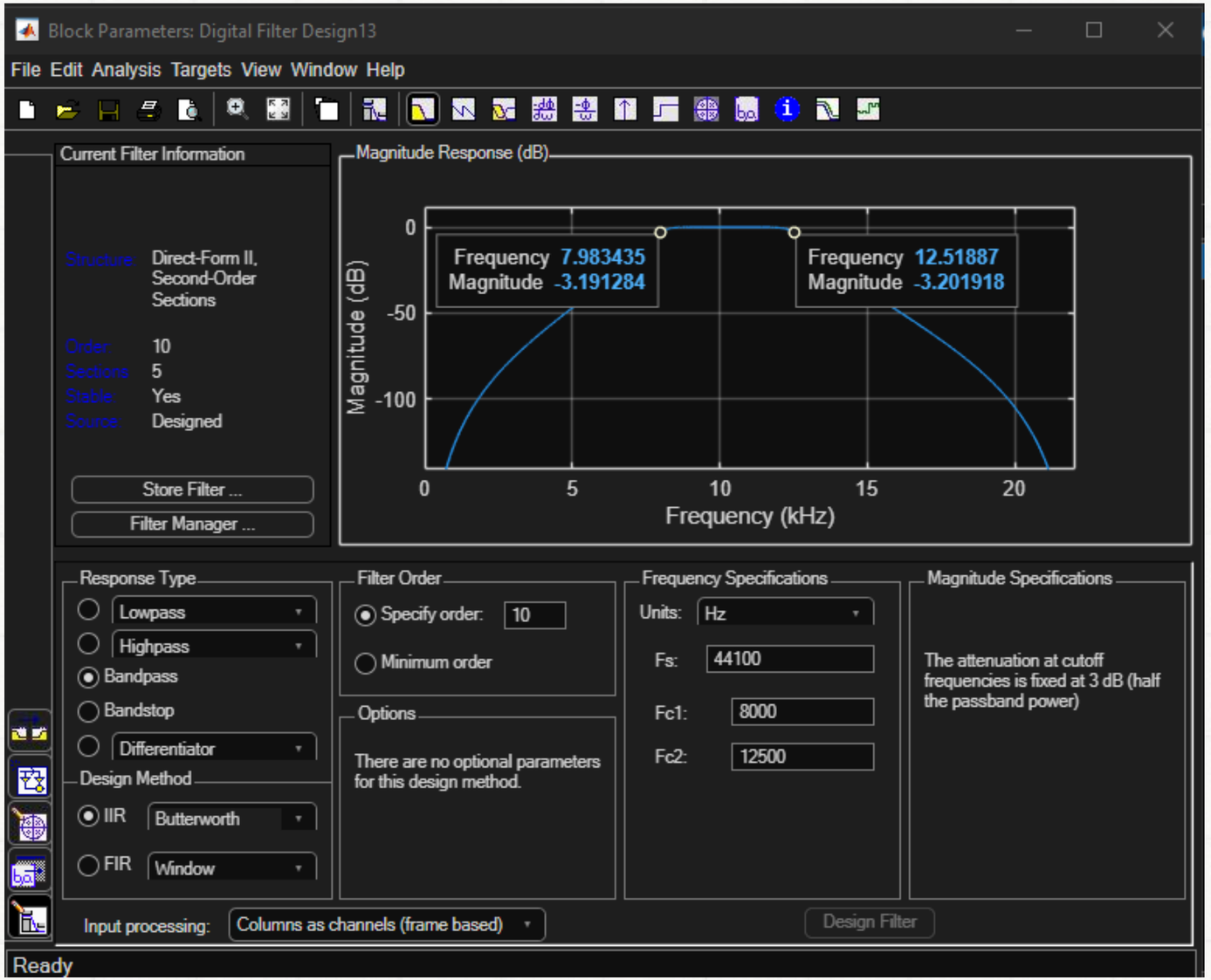
☐ Output end-of-file indicator

Samples per audio channel: 1024

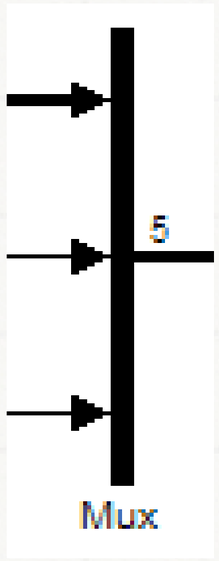
OK Cancel Help Apply

# CONFIGURACIÓN



# CONFIGURACIÓN




Block Parameters: Add

Sum

Add or subtract inputs. Specify one of the following:
a) character vector containing + or - for each input port, | for spacer between ports (e.g. ++|-|++)
b) scalar, >= 1, specifies the number of input ports to be summed.
When there is only one input port, add or subtract elements over all dimensions or one specified dimension.

Main | Signal Attributes

Icon shape: rectangular

List of signs:

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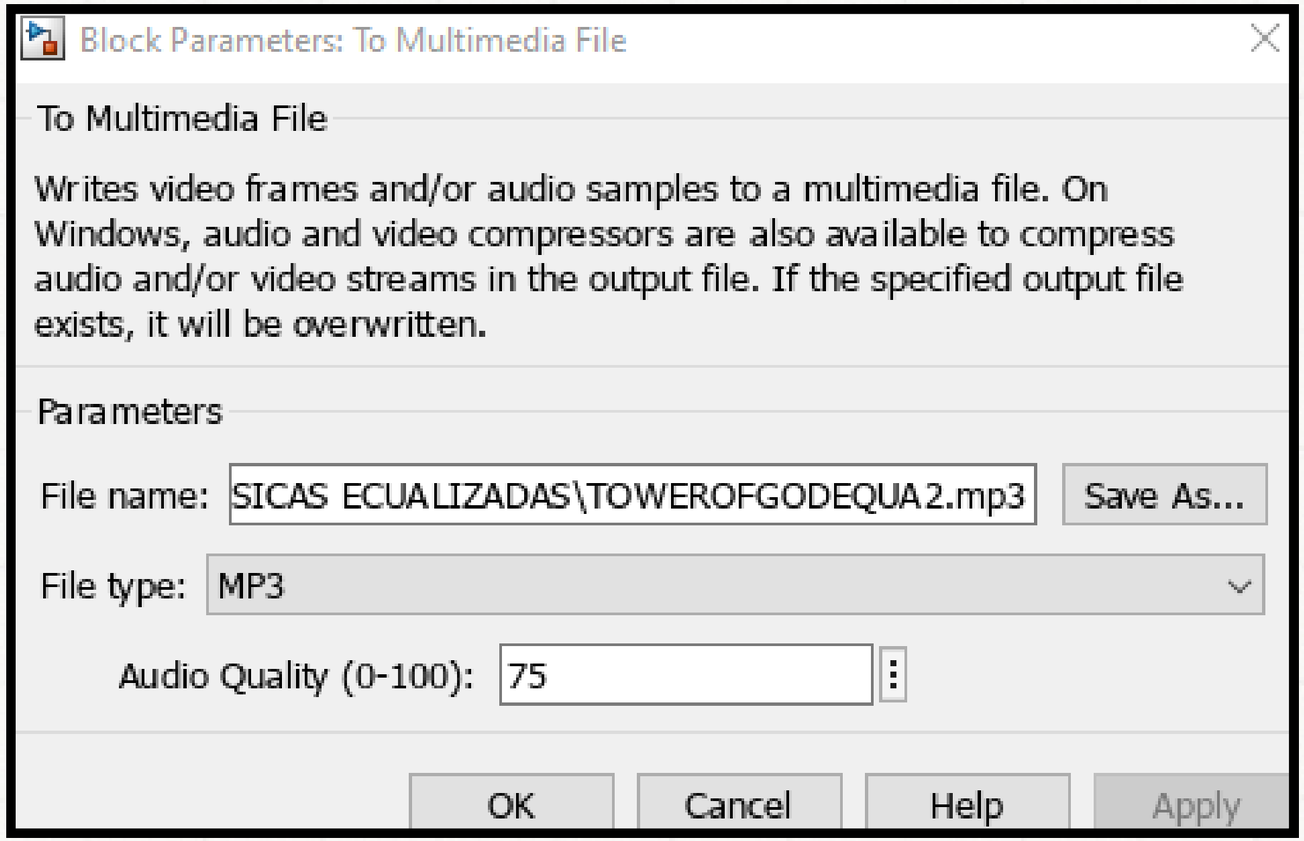
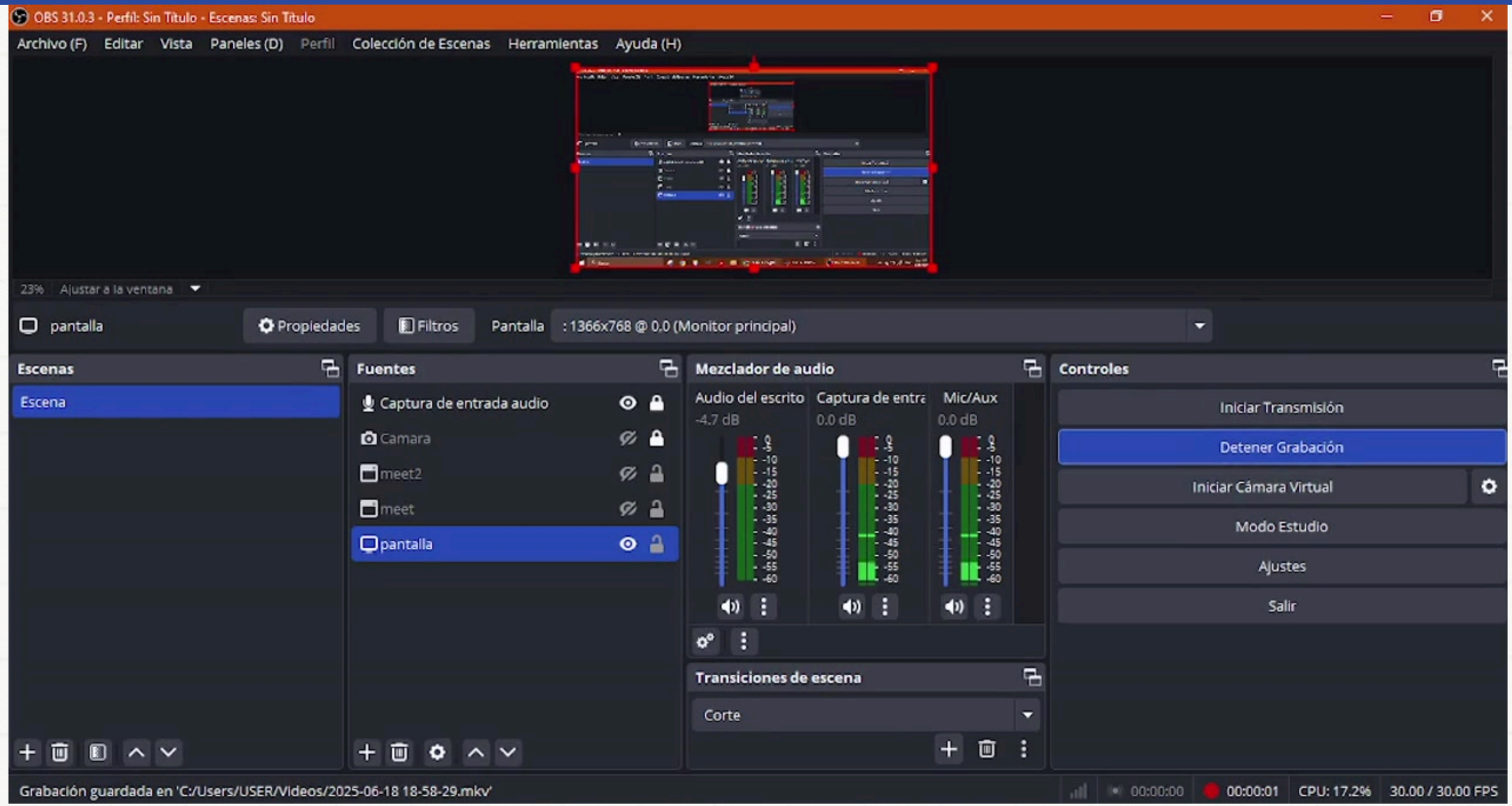
OK | Cancel | Help | Apply

# CONFIGURACIÓN



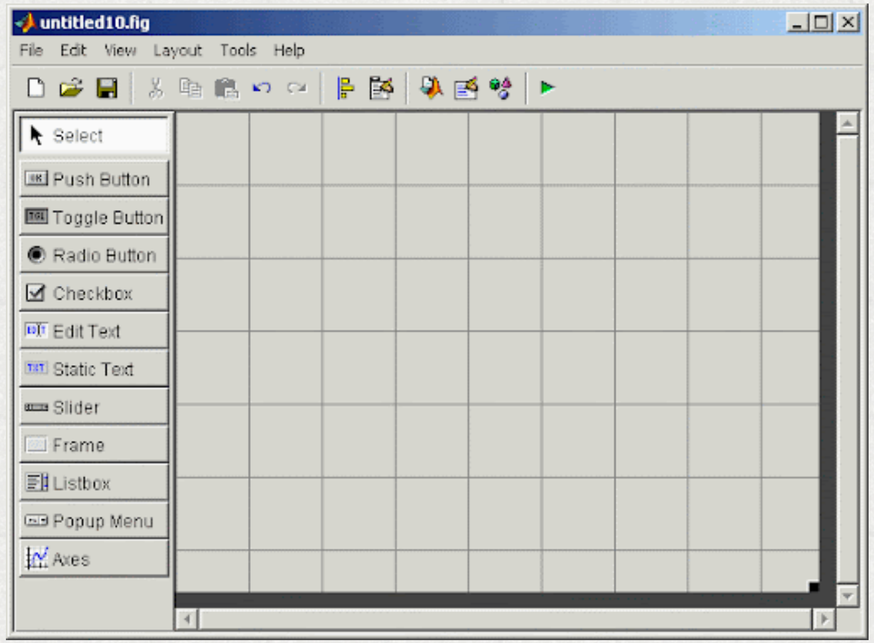
# CODIGO INTERFAZ GUIDE

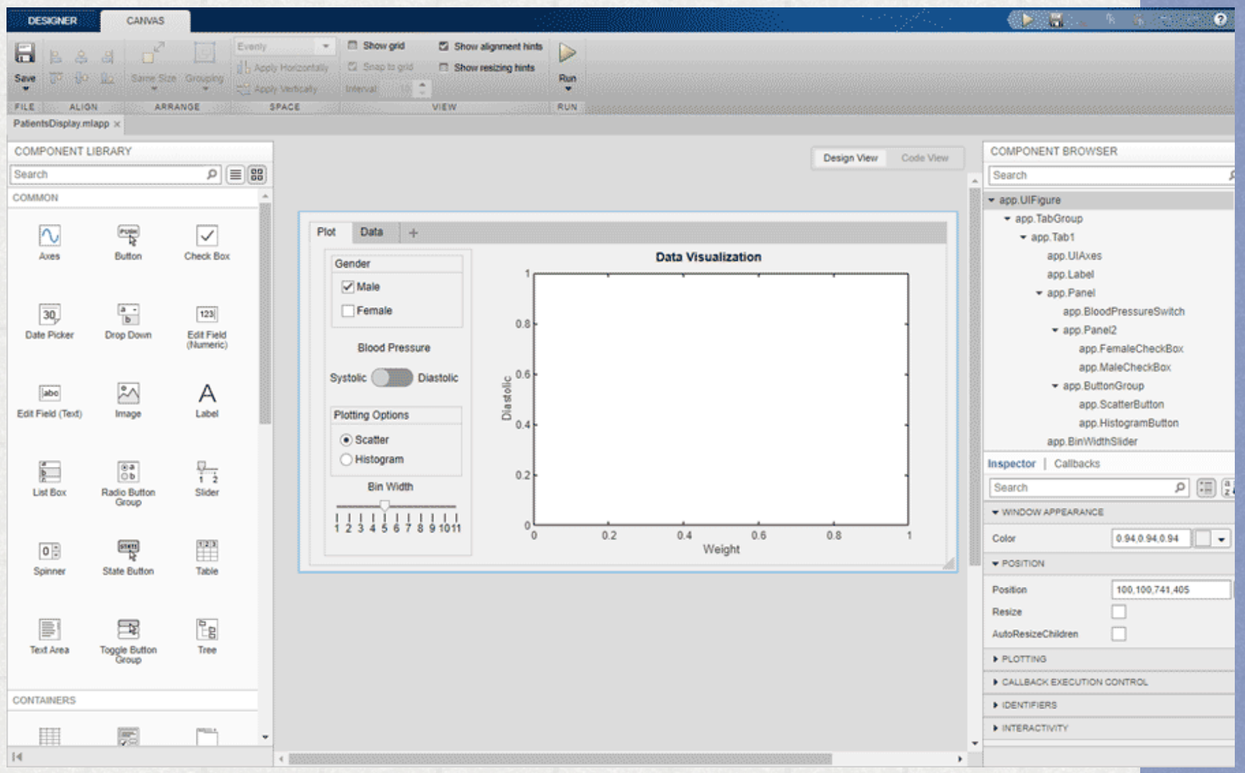


# Interfaz Gráfica

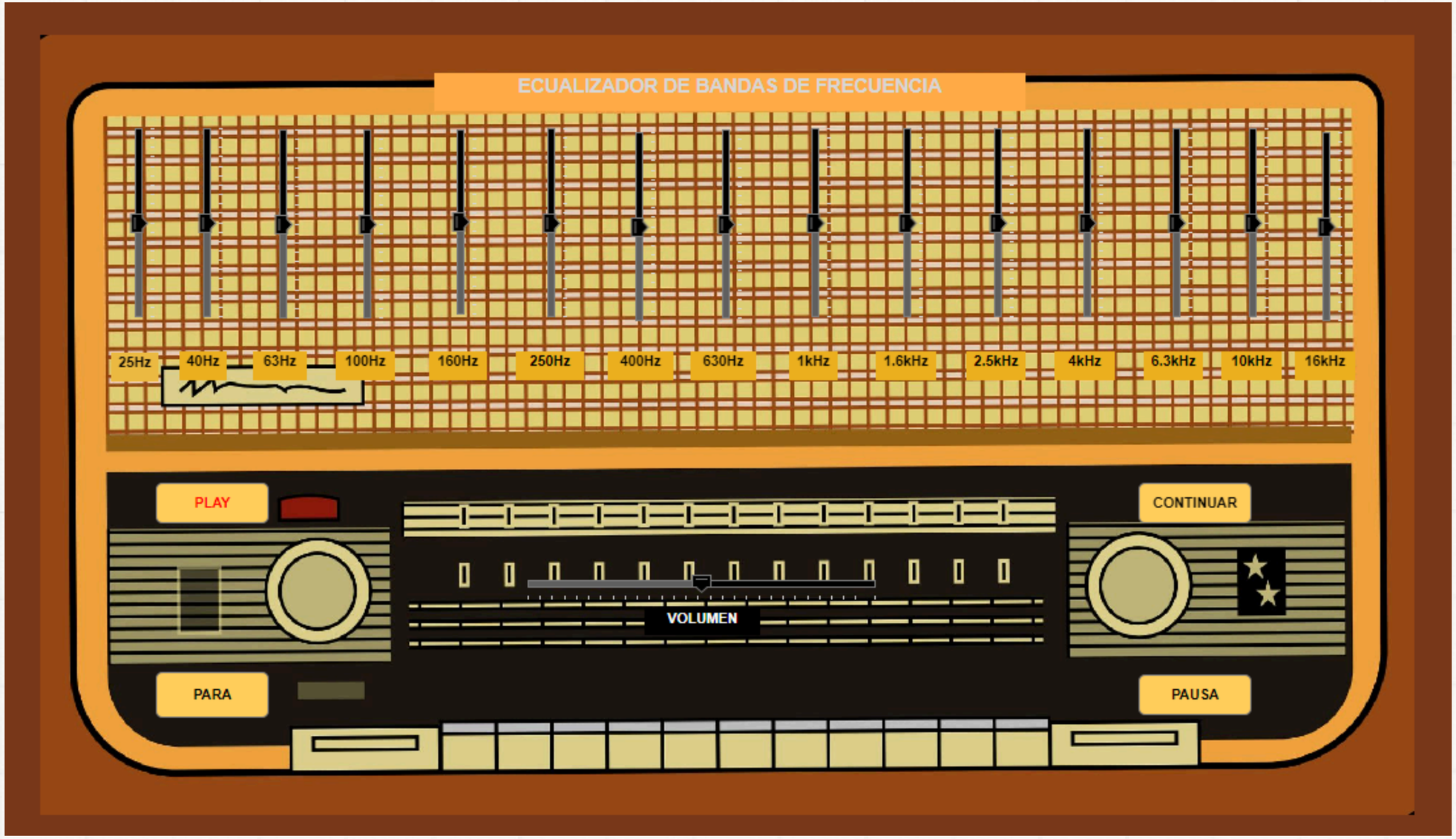
MATLAB Guide

APP DESIGNER

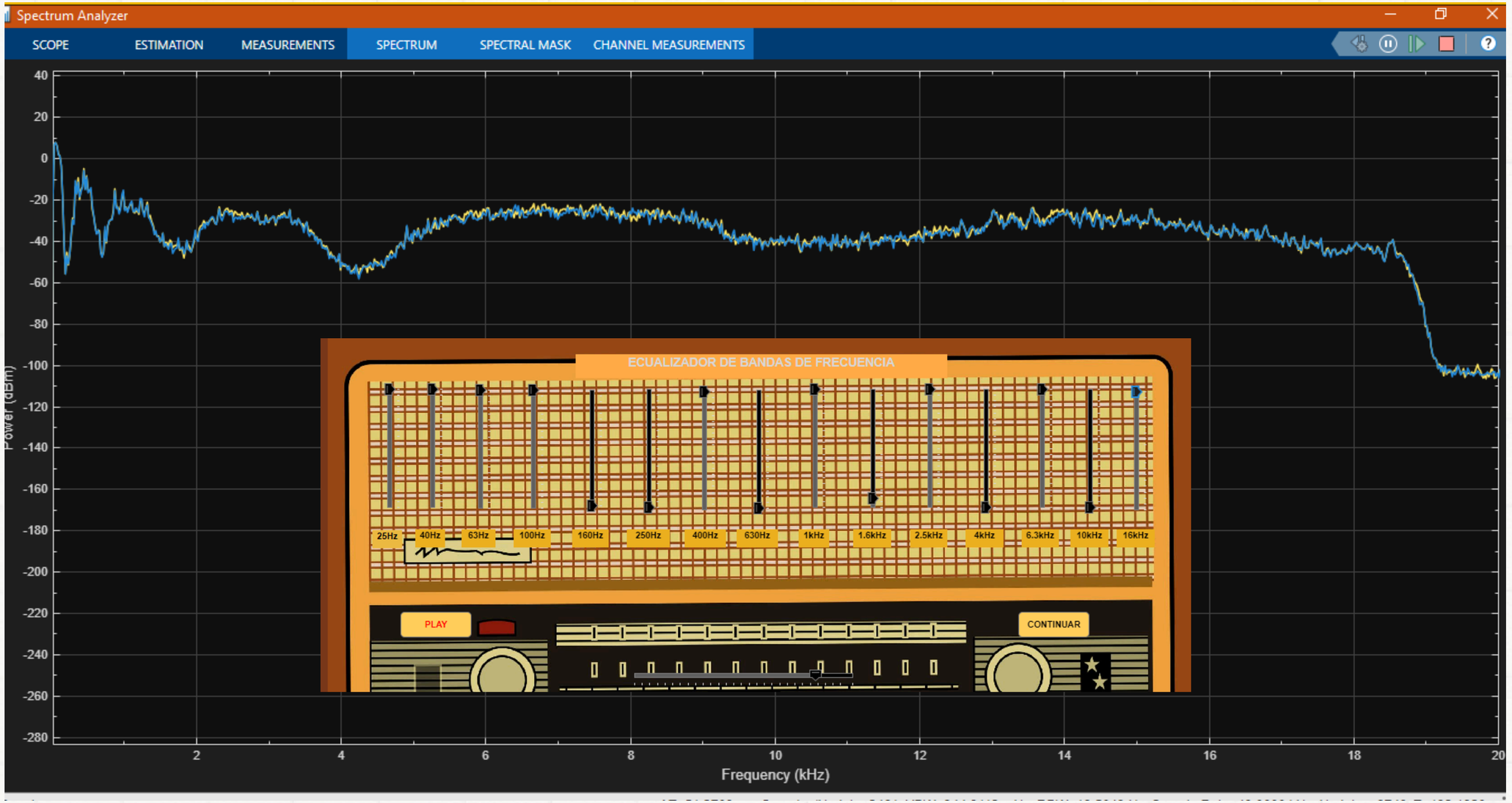


# Interfaz Gráfica



# FUNCIONAMIENTO



# FUNCIONAMIENTO